

ARENA

Human x AI Game Platform

Where humans compete against AI agents in game-theory games
and every game produces actionable behavioral intelligence.

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The Problem

Every AI company building synthetic personas, conversational agents, or AI sales reps faces the same problem: they don't know how well their AI actually mimics real human behavior. They benchmark on surveys, but surveys measure what people SAY, not what they DO.

System	Distribution Accuracy	Gap to Human Ceiling
GPT-5 Mini Personas	61%	-30 points
GPT-5 Personas	62%	-29 points
Gemini 2.5 Flash Personas	64%	-27 points
Gemini 2.5 Pro Personas	67%	-24 points
Artificial Societies	86%	-5 points
Human Replication Ceiling	91%	baseline

Source: Artificial Societies Eval Report, January 2026. Tested on 1,000 UC Berkeley surveys.

These numbers are for stated preferences (surveys). On revealed preferences, behavioral decisions, negotiations, and strategic interactions, the accuracy drops significantly. Nobody has the ground-truth data to measure that gap.

The Solution: Arena

Arena is a platform where humans play competitive game-theory games against AI agents. Players earn money for winning. Companies submit AI agents and game scenarios to benchmark against real humans. Every game produces perfectly structured behavioral data as a natural byproduct of gameplay.

Two Value Propositions

1. Players earn money by winning. Wallet-based KYC payouts. Leaderboard rankings with ELO ratings. The better you play, the more you earn. Games are competitive, fast, and skill-based.

2. Companies test their AI agents. Submit custom AI agents via API. Define game scenarios. Fund reward pools. Get back structured behavioral data, agent performance benchmarks, and distribution accuracy scores against real human baselines.

Why Game Theory Games?

- Constrained choices produce clean, structured data (no open-ended noise)
- Universal: everyone understands money, trust, and fairness (no domain expertise required)
- Decades of published human baselines to compare against
- Players cannot game the system without producing real behavioral data
- The exact benchmarks Stanford used to validate their 1,000-person generative agents
- LLMs are 79% cooperative in Prisoner's Dilemma (much higher than humans): a known, measurable gap

The Games

Game 1: The Split

Based on: Ultimatum Game (the most studied game in behavioral economics)

Two players. A pot of 1000 points. One proposes a split, the other accepts or rejects. Accept: both get the split. Reject: both get zero. Roles alternate. 10 rounds.

Data produced: fairness thresholds, punishment behavior, anchoring effects, adaptation patterns.

Game 2: The Pact

Based on: Iterated Prisoner's Dilemma

Two players. Each round, simultaneously choose: Cooperate or Betray. Both cooperate: +300 each. Both betray: +100 each. Mixed: +500/+0. 15 rounds with the same opponent. Full history visible.

Data produced: cooperation rates, forgiveness patterns, retaliation strategies, end-game effects, strategy classification.

Game 3: The Vault

Based on: Public Goods Game + Free-Rider Detection

4-6 players. Each round, privately invest in a shared vault (multiplied 2x, split equally). Then vote on who free-rode. Correct detection earns bonus. Getting caught costs penalty. 8 rounds.

Data produced: contribution levels, free-rider detection accuracy, punishment effectiveness, cooperation decay.

Game 4: The Auction

Based on: All-Pay Auction / Dollar Auction

3-5 players bid on a 1000-point prize. ALL bidders pay their final bid, not just the winner. Knowing when to STOP is the entire game. 5 auctions per match.

Data produced: escalation of commitment, walk-away thresholds, competitive arousal, sunk cost behavior, learning effects across auctions.

Game 5: The Settlement (2D World)

Based on: Public Goods + Social Deduction in a 2D spatial environment

A top-down pixel-art village. 5 characters (sprites) including the player. Day/night cycle. One character is secretly a saboteur.

Day Phase: Move your sprite around the village. Visit locations (mine, farm, workshop) to gather resources. See where other characters go.

Night Phase: Screen darkens. The saboteur damages a structure or steals from storage.

Morning Phase: Damage revealed. The group sees what happened.

Council Phase: Structured actions: Accuse, Defend, Share Info, Stay Silent. Vote to banish one character.

Data produced: spatial movement patterns, resource allocation decisions, trust calibration, deception detection accuracy, social coalition formation.

Game 6: The Arena (Future)

A 2D marketplace/bazaar. Players are merchants with stalls. AI customers walk through. Players set prices, position stalls, form alliances, sabotage competitors. Full spatial economic simulation.

Research Backing

The Market Gap

No consumer game theory platform exists. Here is the complete landscape:

Platform	What It Is	Monetization
Evolution of Trust (ncase.me)	One-time interactive explainer, viral, single-player	None (Patreon)
MobLab	60+ economics games for university classrooms	\$12/student B2B
oTree / classEx / LIONESS	Academic experiment builders for researchers	Free / institutional
Prolific	Participant recruitment for studies (200K+ users)	Per-response fees
AGI Arena	LLM vs LLM game theory benchmarks	None (research)
prisonersdilemma.io	Blockchain prison-themed game	Crypto token (niche)

Nobody has built a competitive, consumer-facing game theory platform with real rewards, AI opponents, and structured data output.

Stanford Generative Agents

Stanford tested 1,000 AI agents on behavioral economics games (dictator game, prisoner's dilemma, trust game). Agents replicated human survey responses at 85% accuracy. But behavioral accuracy in strategic interactions remains significantly lower. Arena measures exactly this gap.

LLM Behavioral Gaps

- LLMs are ~79% cooperative in Prisoner's Dilemma (humans are ~50-60%)
- Cooperation rates vary by language (higher in English)
- LLMs state beliefs about others' cooperation (~48%) that differ from their own behavior (~79%)
- Anthropic's Persona Selection Model (2026) shows LLMs select from training data personas rather than genuinely modeling human cognition
- Current synthetic persona systems achieve 61-86% on surveys but are untested on revealed preferences

The Synthetic Persona Market

- Artificial Societies (YC W25): \$5.35M seed, 2M+ real-world profiles, 86% distribution accuracy
- Blok: \$7.5M seed, synthetic users for product testing
- Synthetic Society (YC S25): AI synthetic users for bug/UX testing
- Global digital twin market: \$13-16B (2023), projected \$138-195B by 2030
- AI SDR market: \$4.27B (2025), projected \$15B by 2030
- Data labeling market: \$4.89B (2025), growing 28.4% annually

Monetization

Revenue Stream 1: Company Bounties (Primary)

Companies deposit funds to run experiments with their AI agents against real human players. Platform takes 15-20% fee.

Bounty Example	Cost	What Company Gets
500 games of The Split with custom agent	\$2,000	Agent win rate, human response distributions, fairness threshold data
200 games of The Settlement with custom scenario	\$5,000	Spatial behavior data, deception detection patterns, social dynamics
Full behavioral benchmark (all games, 1000 players)	\$10,000	Complete distribution accuracy report across all behavioral dimensions

Revenue Stream 2: Data Licensing

Aggregate, anonymized behavioral distributions sold to:

- Synthetic persona companies (calibration benchmarks)
- AI labs (LLM social behavior benchmarks)
- Behavioral research institutions
- Pricing: per-dataset or subscription

Revenue Stream 3: Platform Fees

- Agent submission and setup fees
- Premium analytics dashboard for companies
- API access for real-time behavioral data

Revenue Stream 4: Player Ecosystem (Future)

- Cosmetic sprite skins for 2D games
- Tournament entry fees with prize pools
- Premium game modes

Player Economics

Metric	Value
Base reward per game	\$3-8 equivalent
Top 10% bonus	+\$5-15 per game
Average game duration	5-15 minutes
Effective hourly rate (top players)	\$20-40/hr
Payout method	Wallet-based (Polygon), KYC verified
Leaderboard	Per-game ELO + global ranking

For context: UserTesting pays \$10-60/session. Surge AI pays \$30-100/hr for expert RLHF work. Sapien pays fractional crypto per labeling task. Arena sits between Sapien (too cheap, too boring) and Surge AI (too expensive, too specialized).

On-Chain Layer (Polygon)

Blockchain is not bolted on for hype. It solves three specific problems:

- 1. Transparent bounty escrow.** Company deposits USDC. Smart contract releases funds on verified game completion. No trust required between anonymous players and companies.
- 2. Portable reputation.** Player score history across games becomes a verifiable on-chain credential. High-reputation players get priority access to premium bounties. Soulbound token or attestation.
- 3. Data provenance.** Each game session is hashed on-chain. Companies can verify: this data came from real games with verified outcomes, not synthetic generation. Anti-model-collapse stamp.

Polygon chosen for: ~\$0.01 gas fees (micropayments viable), mature gaming ecosystem, EVM compatibility, thirdweb SDK support.

Competitive Landscape

Company	What They Do	How Arena Is Different
Artificial Societies	Synthetic persona simulation (86% accuracy on surveys)	Arena provides the ground-truth human data they need to calibrate against
Sapien.io	Gamified data labeling with crypto rewards	Arena has real games (not tagging tasks). Richer behavioral data.
MobLab	Classroom economics games (\$12/student)	Arena is consumer-facing, competitive, with real rewards and AI opponents
Prolific	Participant recruitment for research studies	Arena gamifies participation. Players are engaged, not just paid.
LMSYS Chatbot Arena	Crowdsourced LLM benchmarking (which AI is smarter?)	Arena benchmarks which AI is most human in strategic interactions
Gandalf (Lakera)	Red-teaming game for LLM security	Arena applies the same data-flywheel model to behavioral intelligence

The Pitch

**Arena is where humans compete against AI
in game-theory games, and earn money for winning.**

Companies submit AI agents to test against real humans.
Every game produces structured behavioral intelligence.
The LMSYS Chatbot Arena for human behavior.

The market doesn't exist yet. We're building it.